```
from google.colab import files
uploaded = files.upload()
```

Choose Files Screenshot… 133747.png
Screenshot 2026-08-07 133747.png(image/png) - 142757 bytes, last modified: 8/7/2026 - 100% done
Saving Screenshot 2026-08-07 133747.png to Screenshot 2026-08-07 133747.png

```
import cv2
import matplotlib.pyplot as plt

# Read image
img = cv2.imread("Screenshot 2026-08-07 133747.png")

# Check if image was loaded successfully
if img is None:
    print("Error: Could not load image. Please ensure the image file exists and the path
else:
    img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

    # Create ORB detector
    orb = cv2.ORB_create()

    # Different blur levels
    images = {
        "Original": img,
        "Gaussian Blur": cv2.GaussianBlur(img, (5, 5), 0),
        "Median Blur": cv2.medianBlur(img, 5),
        "Strong Blur": cv2.GaussianBlur(img, (15, 15), 0)
    }

    # Feature detection
    results = []

    for name, image in images.items():
        gray = cv2.cvtColor(image, cv2.COLOR_RGB2GRAY)
```

```
    # Detect ORB features
    keypoints, descriptors = orb.detectAndCompute(gray, None)

    # Draw detected features
    output = cv2.drawKeypoints(
        image, keypoints, None,
        color=(255, 0, 0),
        flags=cv2.DRAW_MATCHES_FLAGS_DRAW_RICH_KEYPOINTS
    )

    results.append((name, output, len(keypoints)))

# Display results
plt.figure(figsize=(12, 8))

for i, (name, output, count) in enumerate(results):
    plt.subplot(2, 2, i + 1)
    plt.imshow(output)
    plt.title(f"{name}\nFeatures Detected: {count}")
    plt.axis("off")

plt.tight_layout()
plt.show()
```


Original
Features Detected: 462
Gaussian Blur
Features Detected: 458